



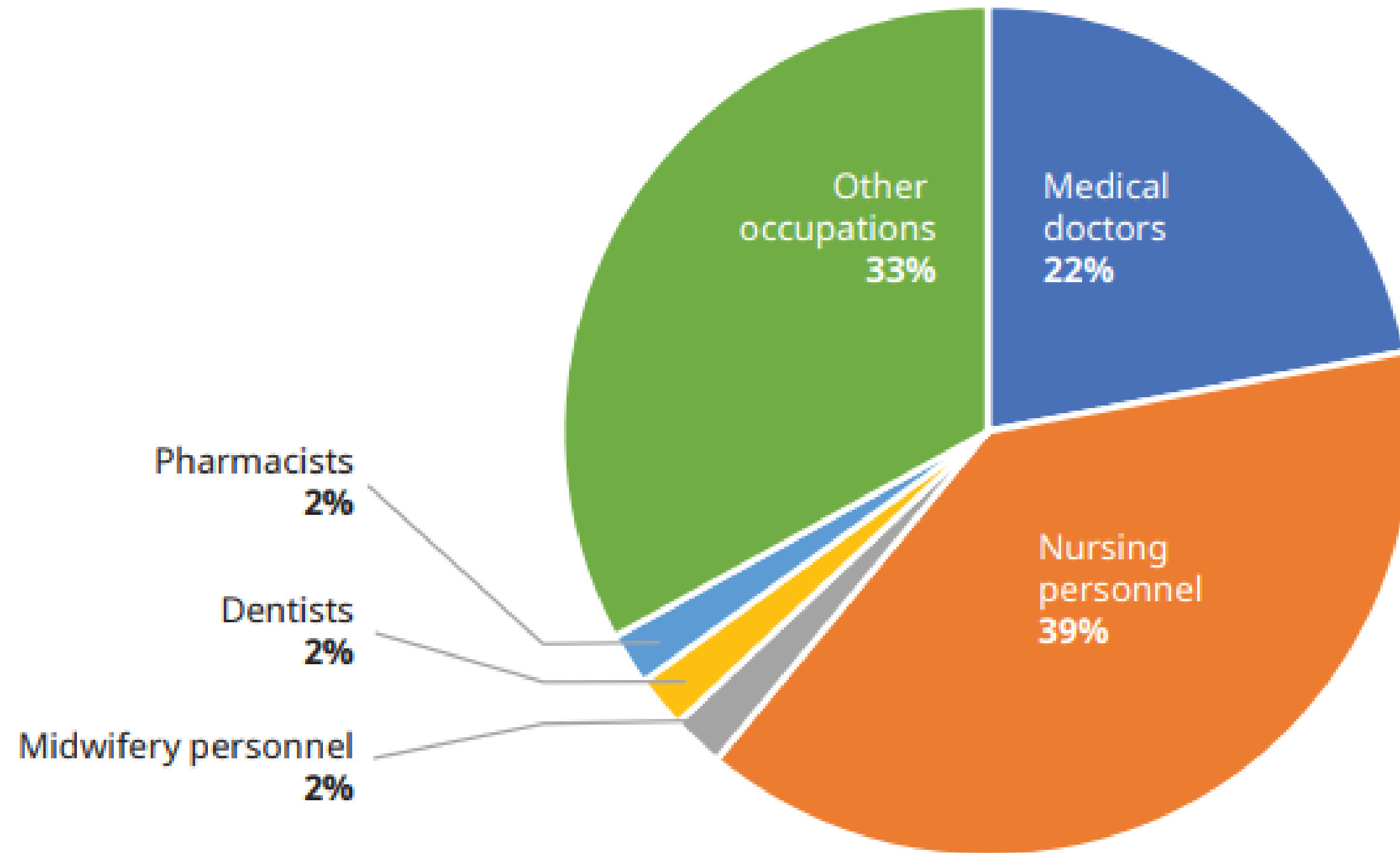
Nursing Workforce: Trend and development strategy



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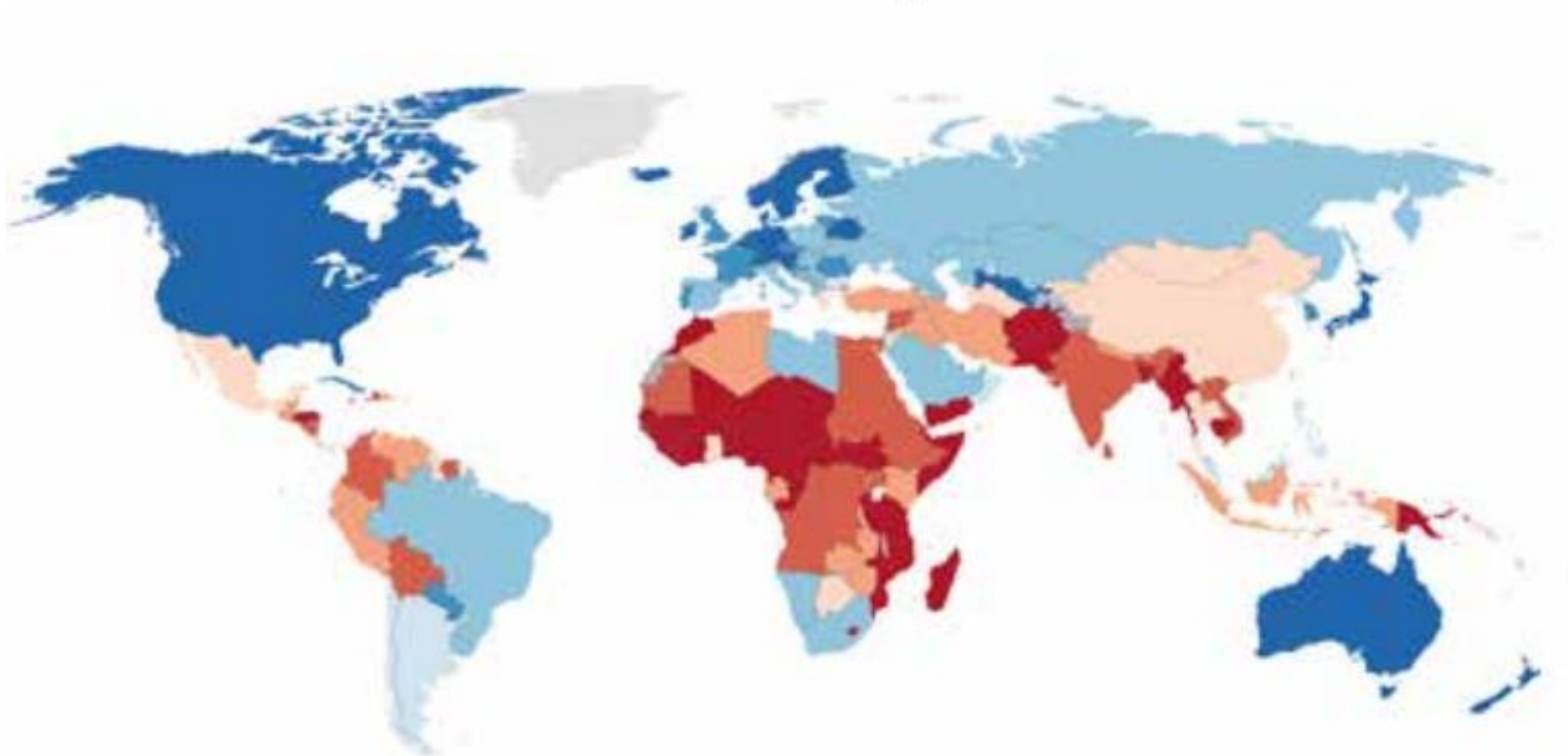
Distribution of the global health workforces



Current situation

- The global stock of nurses is 29.8 million in 2023 and reflects growth from the previous report (27.9 million in 2018)
- The global distribution and density of nurses in 2023 is highly inequitable and it masks a shortage of 5.8 million nurses
- The global density of nurses (37.1 nurses per 10,000 persons)

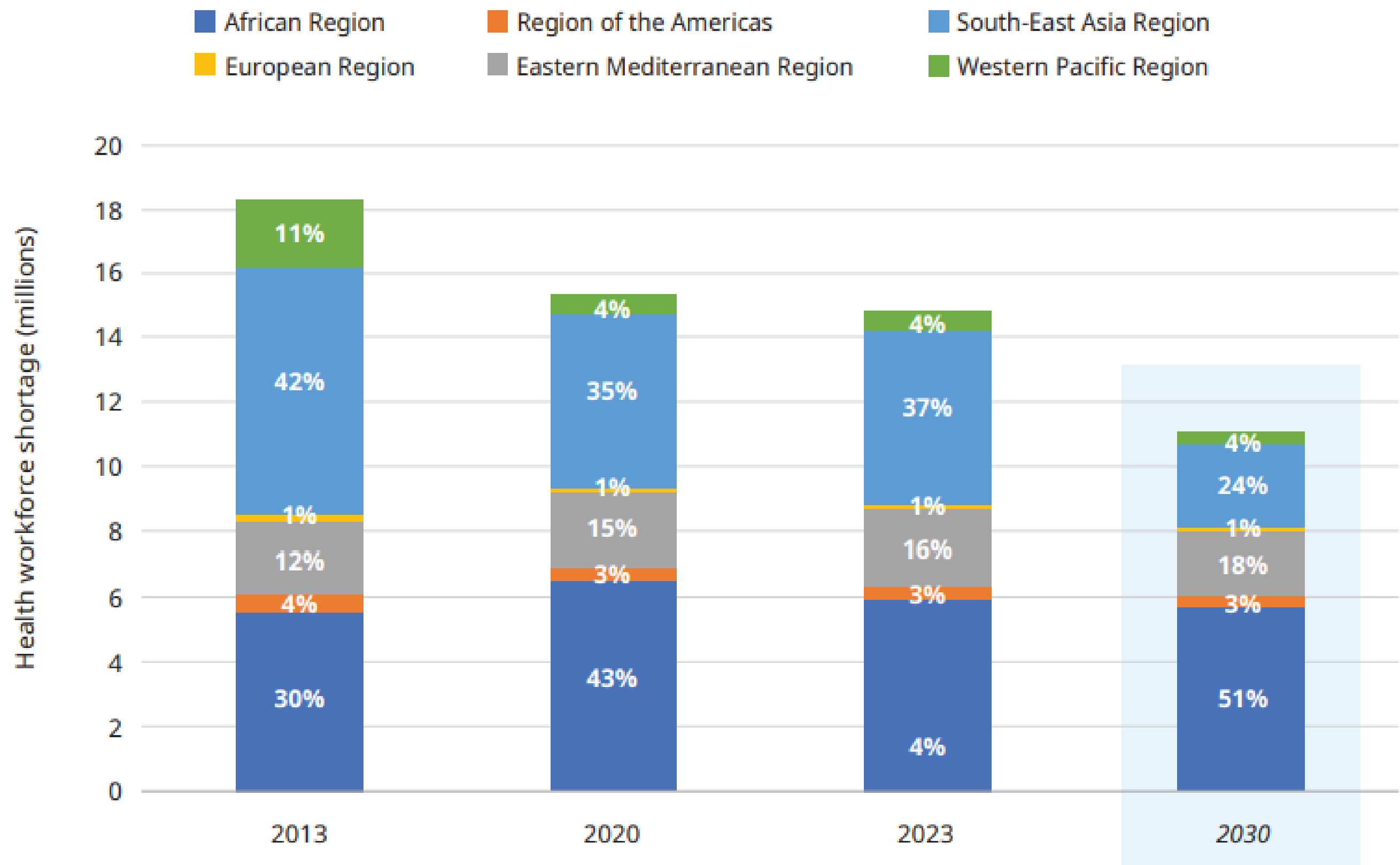
Density of nursing personnel per 10,000 population



Number of nurses globally and density per 10,000 population

BY WHO REGION	Number of countries reporting/total	Number of nursing personnel [million (%)]	Density per 10 000 population [overall (min-max)]
African Region	47/47	1.7 (5.8%)	14.1 (1.2–63.7)
Region of the Americas	35/35	7.5 (25.1%)	72.2 (7.1–133.8)
South-East Asia Region	11/11	3.6 (12.2%)	17.4 (2.0–50.2)
European Region	53/53	7.2 (24.2%)	76.9 (28.0–202.6)
Eastern Mediterranean Region	21/21	1.3 (4.2%)	15.5 (1.2–83.7)
Western Pacific Region	27/27	8.5 (28.5%)	44.0 (5.1–127.1)
BY WORLD BANK INCOME GROUP			
High-income	64/64	13.8 (46.3%)	100.4 (26.5–202.6)
Upper middle-income	54/54	10.3 (34.6%)	36.0 (13.6–103.2)
Lower middle-income	50/50	5.0 (16.9%)	16.3 (1.2–106.8)
Low-income	26/26	0.7 (2.3%)	9.3 (1.2–40.7)
Global	194/194	29.8 (100%)	37.1 (1.2–202.6)

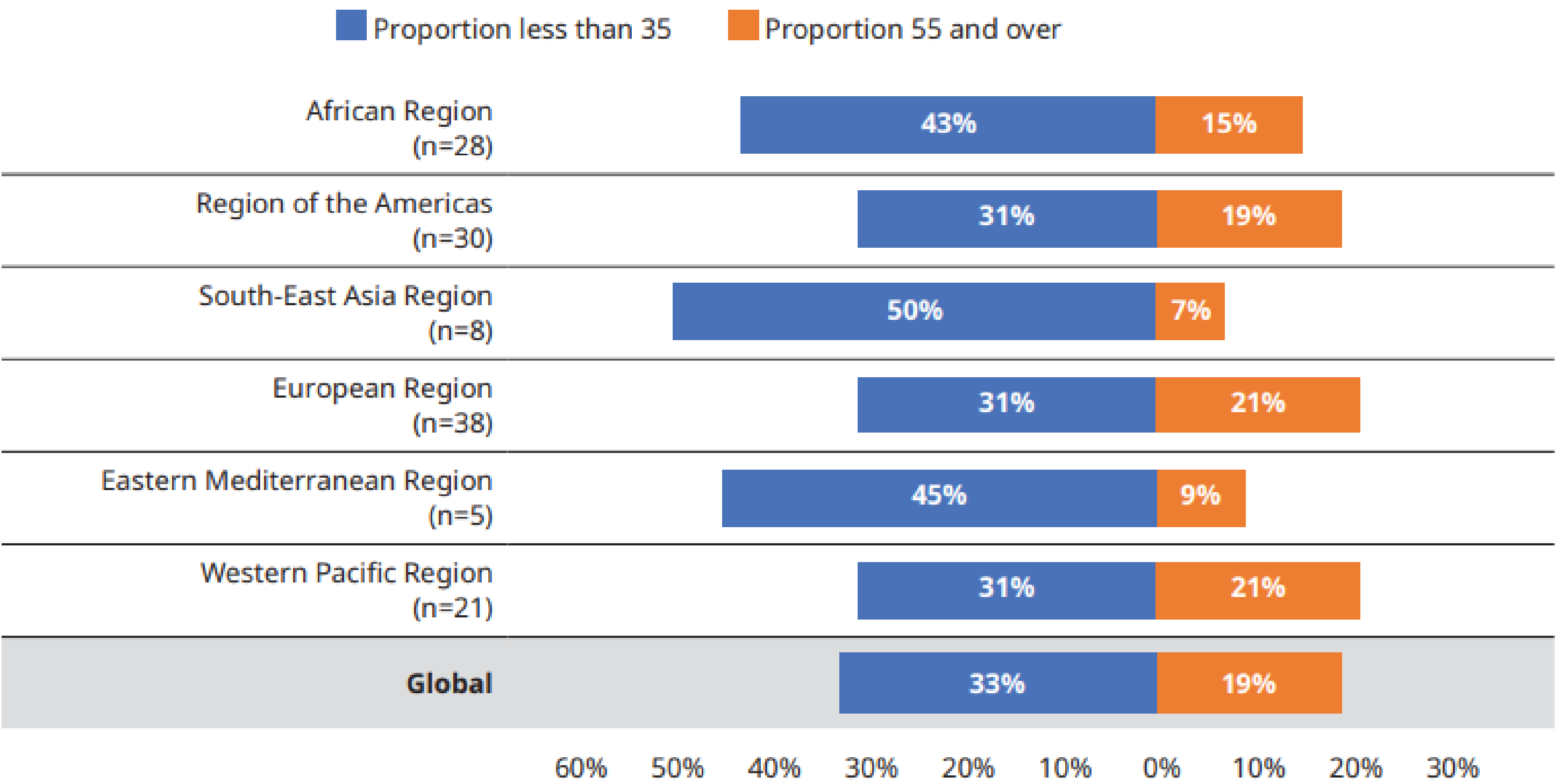
Distribution of the global health workforce shortage



Monthly median entry level wage and salaries

BY WHO REGION	Number of countries reporting/ total	Entry level wage and salary [median (min-max)] (US\$)	Entry level wage and salary [median (min-max)] (PPP\$) ^b
African Region ^a	22/47	480 (150–1384)	1320 (377–3146)
Region of the Americas	23/35	1100 (255–5961)	1827 (442–8920)
South-East Asia Region	7/11	500 (275–953)	1472 (936–2851)
European Region ^a	10/53	2508 (450–5835)	3376 (1060–5368)
Eastern Mediterranean Region ^a	7/21	800 (500–2860)	2629 (1051–5928)
Western Pacific Region ^a	13/27	775 (250–3852)	1641 (527–8413)
BY WORLD BANK INCOME GROUP			
High income ^a	23/64	2302 (915–5961)	3147 (1333–8920)
Upper middle income ^a	25/54	705 (250–1800)	1516 (527–3482)
Lower middle incomes	24/50	539 (231–2077)	1338 (442–8413)
Low income ^a	10/26	360 (150–2556)	945 (377–5928)
Total (n=81)	82/194	774 (150–5961)	1656 (377–8920)

Percentage of nursing personnel aged below 35 years and 55 years or over



Proportion of female and male nursing personnel

WHO REGION	Number of countries reporting/total	Proportion of female nursing personnel	Proportion of male nursing personnel
African Region	34/47	66%	34%
Region of the Americas	31/35	87%	13%
South-East Asia Region	10/11	84%	16%
European Region	39/53	85%	15%
Eastern Mediterranean Region	13/21	73%	27%
Western Pacific Region ^a	23/27	91%	9%
Global	150/194	85%	15%

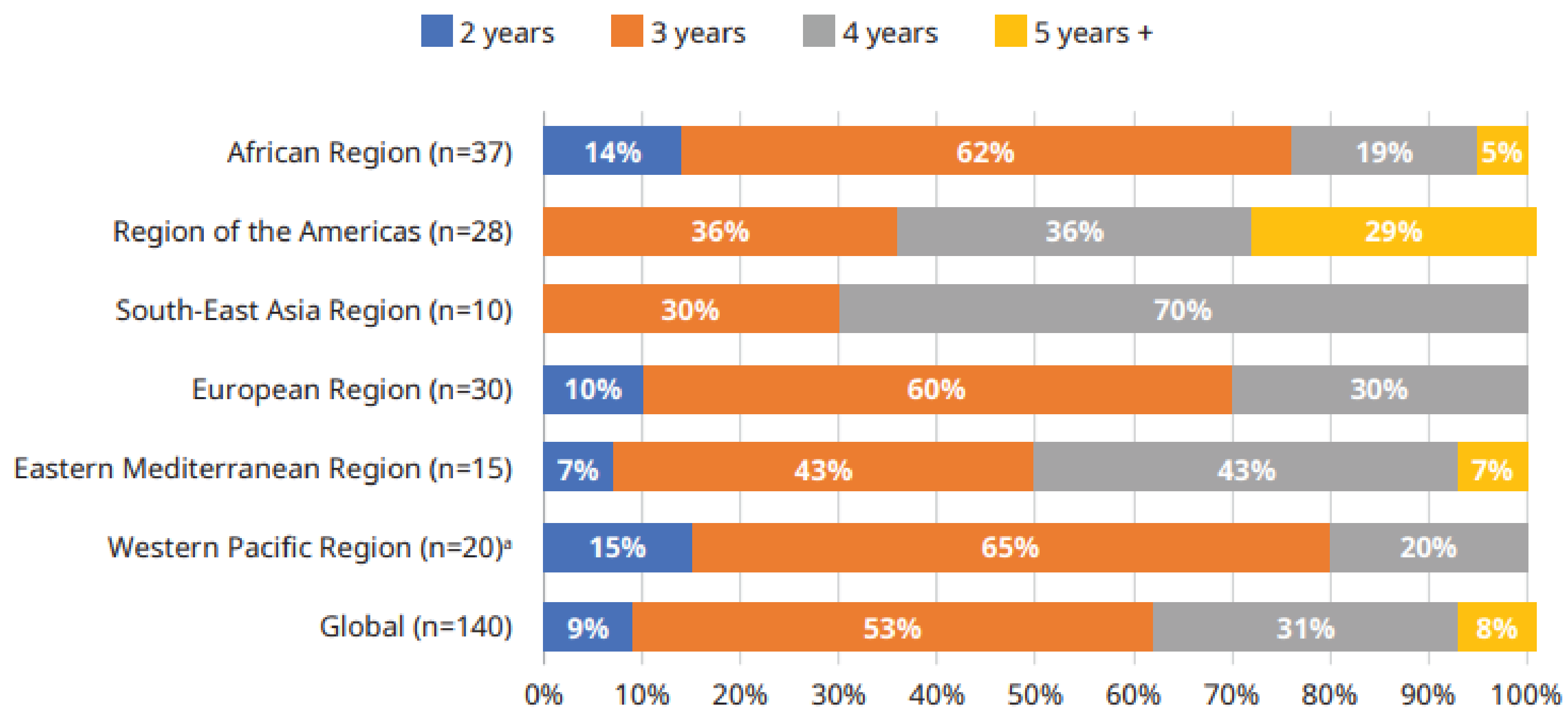
Proportion of foreign-born nursing personnel

BY WHO REGION	Number of countries reporting/total	Proportion of foreign-born nurses
African Region ^a	16/47	3%
Region of the Americas	21/35	6%
South-East Asia Region ^a	5/11	8%
European Region ^a	23/53	14%
Eastern Mediterranean Region ^a	7/21	61%
Western Pacific Region ^a	8/27	26%
BY WORLD BANK INCOME GROUP		
High-income ^a	41/64	23%
Upper middle-income ^a	23/54	8%
Lower middle-income ^a	12/50	1%
Low-income ^a	4/26	3%
Global	80/194	15%

Estimated nursing personnel shortage (in millions)

REGION	Nursing personnel shortage (millions)		Projection
	2020	2023	2030
African Region	2.4	1.9	1.8
Region of the Americas	0.2	0.2	0.2
South-East Asia Region	2.1	2.2	0.8
European Region ^a	–	–	–
Eastern Mediterranean Region	1.2	1.2	1.0
Western Pacific Region	0.2	0.2	0.2
Global	6.2	5.8	4.1

Distribution of countries by duration of education of nursing personnel (in years)





Overview of key issues of Thailand workforce



- WHO has warned that the world needs millions more nurses to achieve UHC, projecting a global shortfall of millions of health workers, with nurses and midwives making up over 50% of this deficit.
- In Thailand, this shortage is not new—the profession has been officially categorized as a "shortage field" by the Cabinet since 1992.
- Over the last 12 months, this systemic problem has been severely intensified by specific national dynamics, directly impacting the stability and capacity of the public health system.





Overview of key issues of Thailand workforce

1. Attrition and Insecurity

- The public sector faces a massive retirements of 19,878 professional (2024–2028).
- High resignation rate among non-permanent new hires, where poor employment conditions (average working 1.2 years) for new graduates in some public hospitals.

2. Demand and Workload

- As Thailand officially became a "complete aged society" in 2022, demand for complex, chronic healthcare has surged. Nurses reportedly working over 60 hours per week on average, compromising both their well-being and patient safety.

3. Primary Care Deficit

- The shortages are most acute in the foundational healthcare system. There is a critical deficit of 12,633 professional nurses required in the primary care system.





Systemic Employment Instability

- **Job Precarity:** New graduates are often hired at a base wage as low as 12,000 THB/month and face a protracted pathway, sometimes requiring more than five years, to achieve stable permanent (civil servant) status.
 - **The Mobility Driver:** This job and wage uncertainty is the leading cause of high turnover and fuels the internal push factor toward the private sector.
 - **Compensation Inequity:** Beyond the base wage, the gap in supplementary pay between professional nurses and medical doctors has been observed to widen to as much as 26 times greater for doctors. This extreme disparity severely impacts morale and retention.
- 

Strategic Advocacy: Demanding Structural Change

- Primary Advocacy Goal: To eliminate the deeply ingrained issue of non-permanent employment by achieving structural and policy change.
- Key Action: The Nurses' Association and Nursing Council (NAT/TNMC) are in continuous negotiation with the government (MoPH and OCSC).
- Negotiation Focus: The focus is on ensuring that sufficient permanent Civil Servant positions are allocated and prepared for the massive influx of new nursing graduates currently being produced.
- Impact: This action is critical to guarantee immediate employment stability, fair welfare, and retention for the entire public nursing workforce, moving Thailand from a production success to a retention success.



A Workforce Defined by Scarcity and Imbalance

Metric	Detail	Implication
Total Licensed Nurses	250,582 (As of Jan 2024)	Represents the total potential supply pool.
Actively Practicing Nurses	Approximately 175,407 (70% of licensed total)	Indicates that nearly one-third of licensed nurses are not in active practice.
National Ratio (2022)	1:371 (1 nurse per 371 people)	Current supply ratio.
National Target Ratio (2027)	1:326 (1 nurse per 326 people)	The target ratio set by the national strategy, requiring the production of over 15,000 additional nurses.



Geographical Disparity and Mobility

Region	Nurse-to-Population Ratio (1:X)	Comparative Note
Bangkok	1:143	Best Ratio (Highest concentration and resource availability).
Central Region	1:373	Close to national average.
Northern Region	1:397	Below national average.
Southern Region	1:369	Close to national average.
Eastern Region	1:333	Better than national average.
Northeast Region	1:584	Worst Ratio (The most severe shortage and greatest need).

Retention Crisis: The Cost of Instability

- **Core Message:** The challenge is not the workload, but the structural employment insecurity that drives nurses out of the public sector almost immediately.

Metric	Result	Implication
Resignation Rate	40.84% of non-permanent nurses resign overall.	Nearly two out of every five non-permanent hires are lost.
First-Year Turnover	48.68% of non-permanent staff resign during their first year.	Turnover occurs alarmingly fast, draining training resources immediately.
Average Tenure	1.2 years for new graduates who fail to secure permanent status.	The constant high churn prevents professional continuity and service quality.

- **Career Stagnation:** Experienced nurses often face rigid promotional structures, remaining at the Senior Professional Nurse level well into their mid-40s, reducing incentives for long-term service.

Negotiation: Securing the Investment in Production



The Project: "Project to Increase Nursing Personnel Production (Academic Years 2023–2027)."



The Scale: A national commitment involving 62 institutions, aiming to produce 15,985 additional nurses.



The Investment: Securing a substantial budget of 7,033.40 million THB for this production effort.

Forward Strategy: Warm Community Nursing Clinic

- The “Warm Community Nursing Clinic” is a network of Nursing and Midwifery Clinics that serves as a participating service unit under the National Health Security Office (NHSO) of Thailand.
- Purpose: The clinics aim to increase accessibility to primary healthcare and reduce patient congestion in larger hospitals.
- Home Health Care: Providing care for specific groups of patients at home, such as:
 - Patients with chronic diseases whose symptoms are controlled or mildly disabled.
 - Patients requiring wound care, tube feeding, or catheter care.
- To bolster healthcare accessibility, the Nurses' Association of Thailand (NAT's) strategically encouraged retired and freelance nurses to leverage their valuable experience and knowledge by joining the community clinics



Forward Strategy: Executing the Transition to Retention

- **Intensive Advocacy for Permanent Positions (Retention)**
 - Action: Maintain core negotiation with the government to ensure the preparation and allocation of permanent Civil Servant positions.
 - Rationale: Immediately stop the 40.84% early-career turnover and protect the national investment.
- **Strategies for Equitable Distribution (Deployment)**
 - Action: Develop and implement strategies to ensure the new workforce is equitably distributed across the country's 13 health regions.
 - Goal: Actively counteract internal migration and reduce severe disparities, such as the alarming contrast between Bangkok (1:143) and the Northeast Region (1:584).

National Long-Term Vision Meets Global Goals

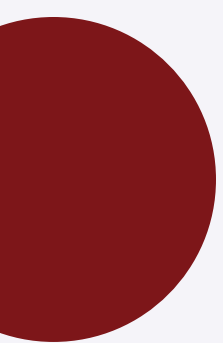
- **National Integration (The 20-Year Framework)**

- Alignment: The nursing workforce development strategy is fully integrated into the National Strategy (2017–2036) and the National Health System Reform Plan (2022).
- Impact: This ensures long-term government commitment and views nursing human resources as an essential asset for achieving comprehensive national development goals.

- **Global Alignment (WHO & UN Mandates)**

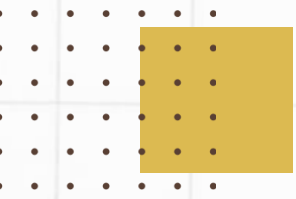
- International Consistency: The national strategy is explicitly developed to be consistent with the direction of the Global Nursing and Midwifery Strategy.
- Key Goals: It directly supports the Sustainable Development Goals (SDG 3) (ensuring healthy lives and well-being for all) and is considered essential for Thailand to achieve and maintain Universal Health Coverage (UHC).





Integrated into National and Global Vision

Strategic Element	Goal
Health Manpower Reform	Recognized as one of the ten core reform priorities in the National Health System Reform Plan (2022).
Primary Care Uplift	Actively promoting accessible basic services to reduce the unsustainable burden on central hospitals, requiring a strong primary care nurse workforce.
Digital Integration	Mandates connecting all health services and data, necessitating a digitally-competent nursing workforce



Integrated into National and Global Vision

A 3-Pillar Skill Strategy for our nurses

Upskill	Reskill	Newskill
Enhance clinical expertise (like advanced wound care).	Adapt existing skills for remote and digital settings (like telehealth and motivational interviewing) to become effective Case Managers	Introduce critical administrative and coordination competencies to transform them into strategic Care Managers.

Key outcomes:

- 1.**Quality Improvement:** Reduced complications and a significant decrease in costly hospital readmission rates
- 2.**Cost Saving:** Reducing unnecessary treatment, treatment burden, and redundant visits to emergency departments



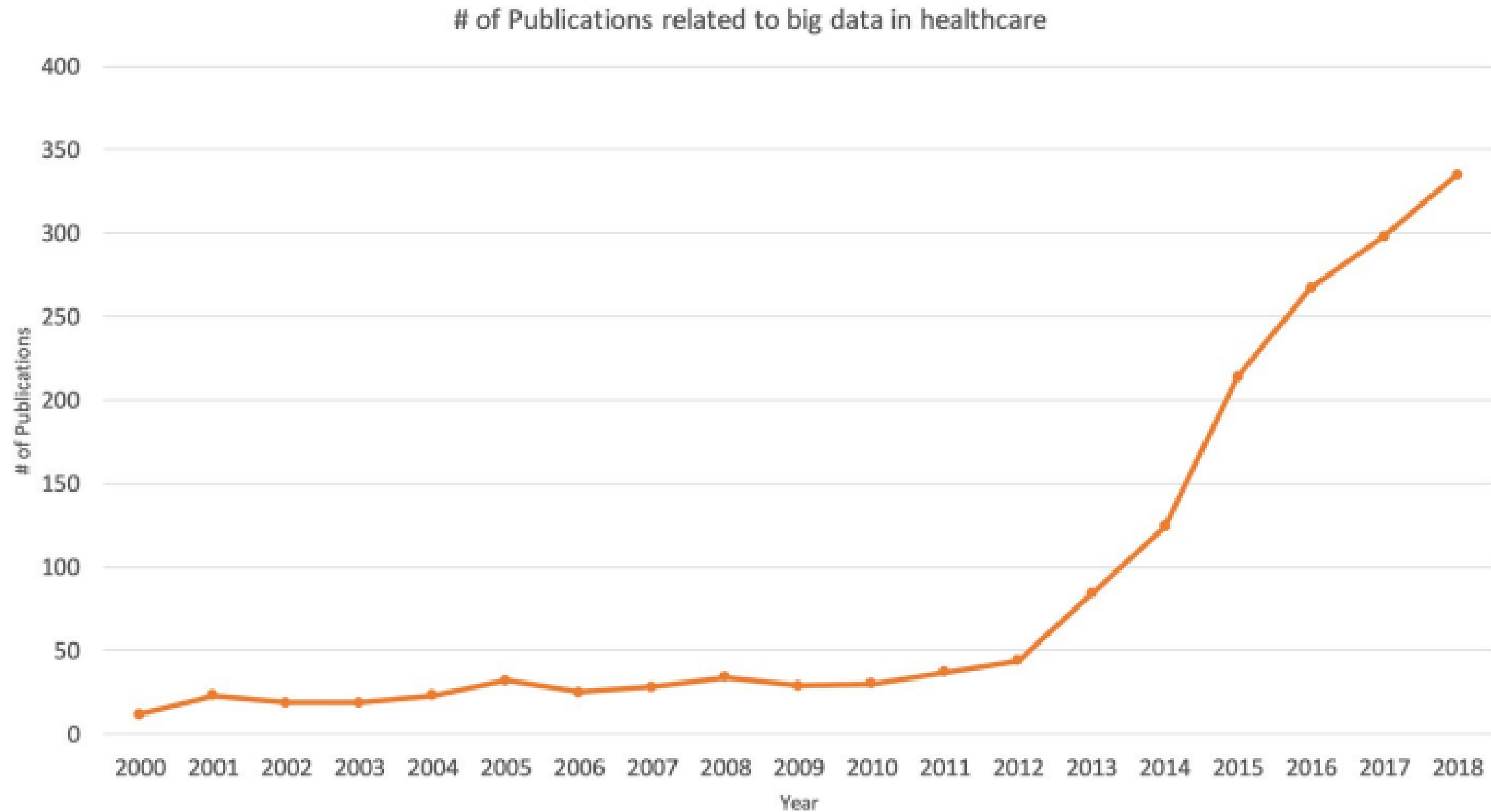
“Big Data is volume, high-velocity and high-variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making”

Gartner, 2012





Big Data in healthcare



Benefits of Big Data



1. Increasing earlier diagnosis and the effectiveness and quality of treatments
 - Discovery of early signals and disease intervention, reduced probability of adverse reactions, etc.
2. Widening possibilities for monitoring of diseases
 - Capturing, manipulating, and analyzing massive datasets



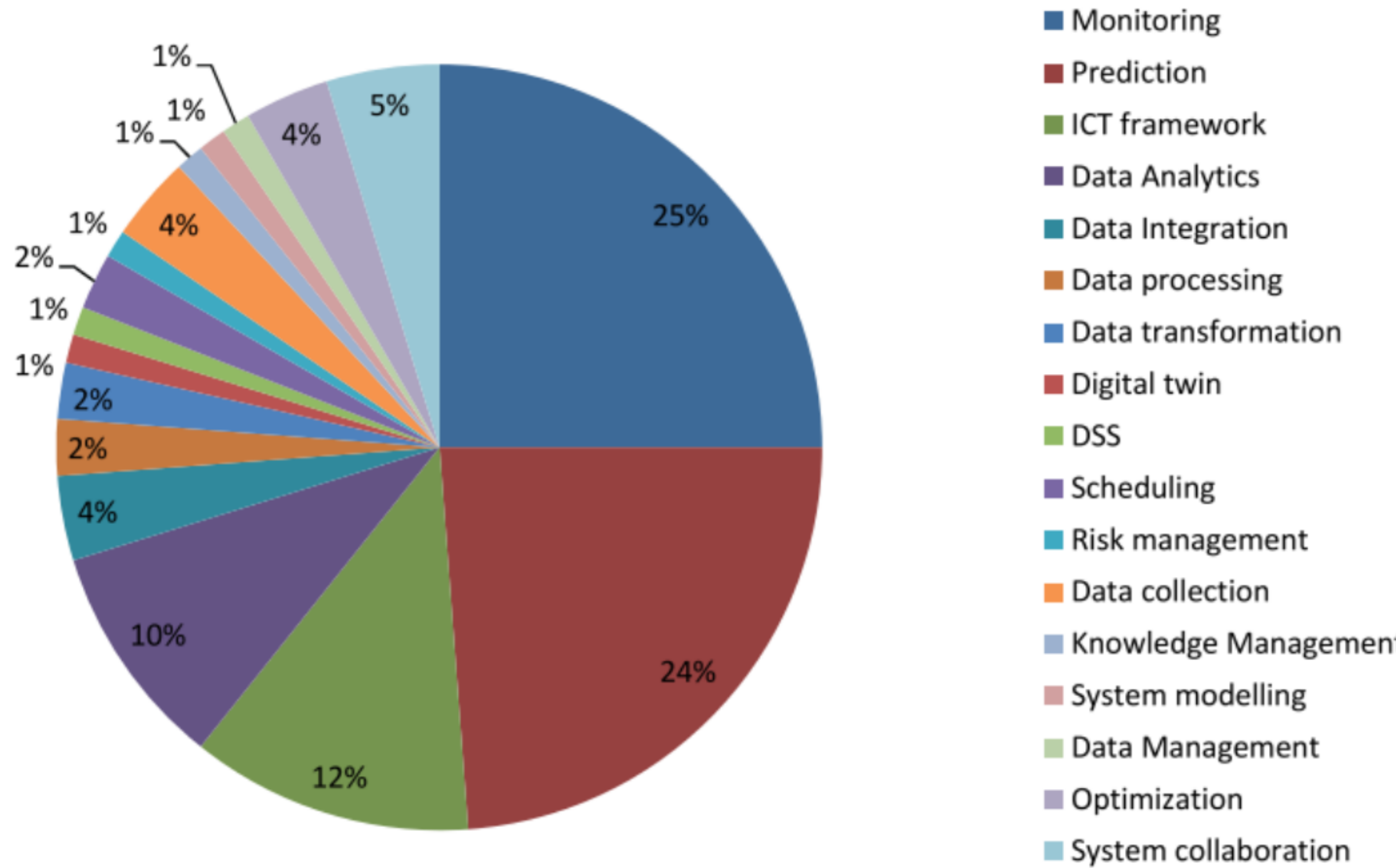
Benefits of Big Data



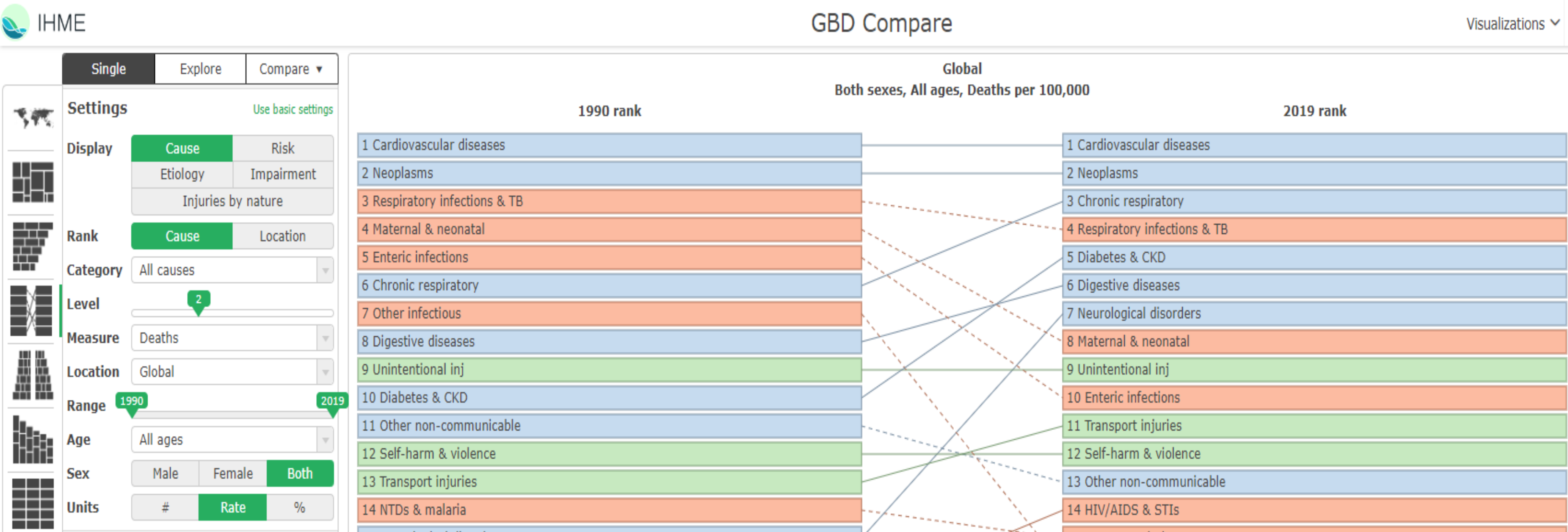
3. Improvement of patient safety
 - Ability to make more informed medical decisions based on directly delivered information to the patients
4. Prediction of outcomes
 - Yield new insights into risk factors that lead to disease



Allocation of Big Data applications



Example of Big Data Monitoring



What risk factors drive the most death and disability combined?

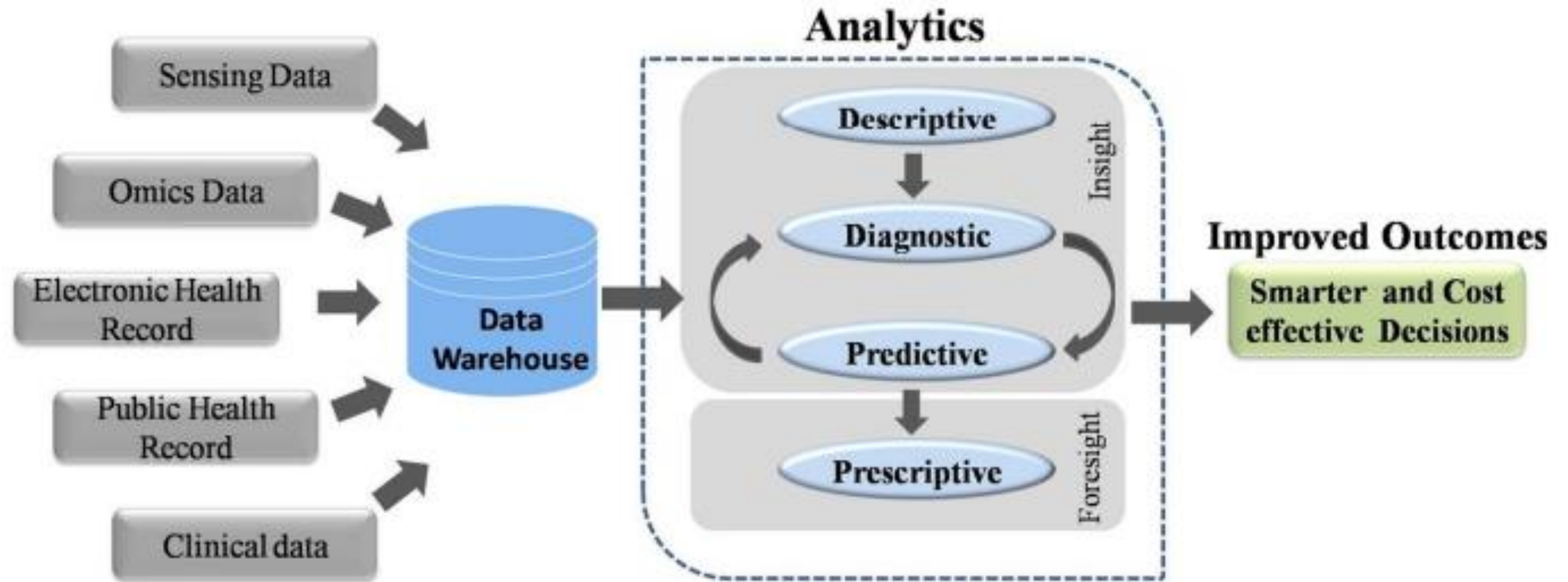
Metabolic risks

Environmental/occupational risks

Behavioral risks

Risk	2009 rank	2019 rank	Change in DALYs per 100k, 2009–2019
Tobacco	1	1	↑ +340.9
High fasting plasma glucose	4	2	↑ +897.7
High body-mass index	7	3	↑ +787.8
High blood pressure	2	4	↑ +374.3
Alcohol use	3	5	↑ +191.0
Dietary risks	6	6	↑ +316.5
Air pollution	5	7	↓ -46.0
Kidney dysfunction	8	8	↑ +318.2
Occupational risks	9	9	↓ -19.3
Unsafe sex	10	10	↑ +18.4

Big Data analytics



Example of Big Data Prediction



ASCVD Risk Estimator Plus

Estimate Risk

Therapy Impact

Advice

App should be used for primary prevention patients (those without ASCVD) only.

Current Age ⓘ *

Age must be between 20-79

Sex *

Male

Female

Race *

White

African American

Other

Systolic Blood Pressure (mm Hg) *

Value must be between 90-200

Diastolic Blood Pressure (mm Hg) *

Value must be between 60-130

Total Cholesterol (mg/dL) *

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

Value must be between 20 - 100

LDL Cholesterol (mg/dL) ⓘ ○

Value must be between 30-300

History of Diabetes? *

Yes

No

Smoker? ⓘ *

Current ⓘ

Former ⓘ

Never ⓘ

On Hypertension Treatment? *

Yes

No

On a Statin? ⓘ ○

Yes

No

On Aspirin Therapy? ⓘ ○

Yes

No

Do you want to refine current risk estimation using data from a previous visit? ⓘ ○

Yes

No

<https://tools.acc.org/ascvd-risk-estimator-plus/#!/calculate/estimate/>

Example of Big Data Prediction

AMERICAN COLLEGE of CARDIOLOGY

ASCVD Risk Estimator Plus

Estimate Risk

Therapy Impact

Advice

23.7%
High

Current 10-Year
ASCVD Risk**

Lifetime ASCVD Risk: 69%

Optimal ASCVD Risk: 2.3%

Unit of Measure

US

SI

Reset All

App should be used for primary prevention patients (those without ASCVD) only.

Current Age ⓘ *

40

Age must be between 20-79

Sex *

✓ Male

Female

Race *

White

✓ African American

Other

Systolic Blood Pressure (mm Hg) *

150

Value must be between 90-200

Diastolic Blood Pressure (mm Hg) *

80

Value must be between 60-130

Total Cholesterol (mg/dL) *

250

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

40

Value must be between 20 - 100

LDL Cholesterol (mg/dL) ⓘ ○

140

Value must be between 30-300

History of Diabetes? *

✓ Yes

No

Smoker? ⓘ *

✓ Current ⓘ

Former ⓘ

Never ⓘ

On Hypertension Treatment? *

✓ Yes

No

On a Statin? ⓘ ○

✓ Yes

No

On Aspirin Therapy? ⓘ ○

Yes

✓ No

<https://tools.acc.org/ascvd-risk-estimator-plus/#!/calculate/estimate/>

Example of Big Data Prediction



ASCVD Risk Estimator Plus

Estimate Risk

Therapy Impact

Advice

23.7%
High
Current 10-Year
ASCVD Risk**

Lifetime ASCVD Risk: **69%** Optimal ASCVD Risk: **2.3%**

Project Risk Reduction by Therapy

[Reset](#)

[View Advice Summary for this Patient](#)

Projected 10-Year ASCVD Risk

17.3% with Smoking Cessation



Quit
Smoking ⓘ



Start/Intensify Statin ⓘ



Start/Add Blood Pressure
Medication(s) ⓘ



Start/continue aspirin
therapy ⓘ

<https://tools.acc.org/ascvd-risk-estimator-plus/#!/calculate/estimat>

RISK REDUCTION

CVDs risk 23.7%



If quit smoking, the projected risk is 17.3%



If start statin, the projected risk is 17.8%



If control blood pressure, the projected risk is 17.4%



If continue aspirin therapy, the projected risk is 21.4%



If quit smoking, start statin, control blood pressure, and continue aspirin therapy, the projected risk is 8.6%

Example of Big Data Prediction



ASCVD Risk Estimator Plus

Estimate Risk

Therapy Impact

Advice

Visit Summary

Below is a summary of patient’s risk, treatment options, and treatment advice based on the data provided.

Email Advice

Print with Complete Treatment Advice

Print

Treatment Advice*

[Expand All](#)

- ▶ LDL-C Management (for this Patient)
- ▶ Blood Pressure Management (for this Patient)
- ▶ Tobacco Cessation (for this Patient)
- ▶ Diabetes Mellitus Management (General)
- ▶ Lifestyle Recommendations (General)
- ▶ Aspirin Use Recommendations (for this Patient)
- ▶ Immunization Practice (General)
- ▶ Therapy Safety Information (General)



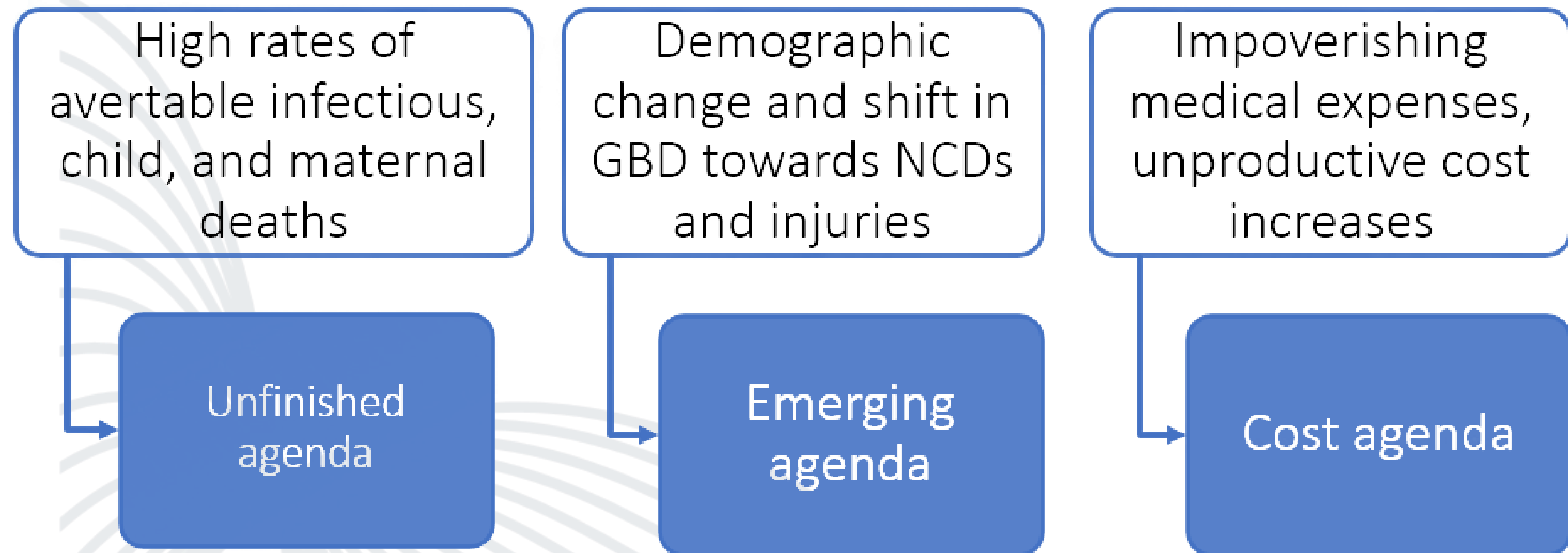
Unique Features of Big Data in Healthcare

- Classic
 - Volume
 - Variety
 - Velocity
 - Variability
 - Veracity
- Healthcare component
 - Heterogeneity
 - Incompleteness
 - Timeliness and longevity,
 - Data privacy
 - Ownership

Number of big data papers per discipline

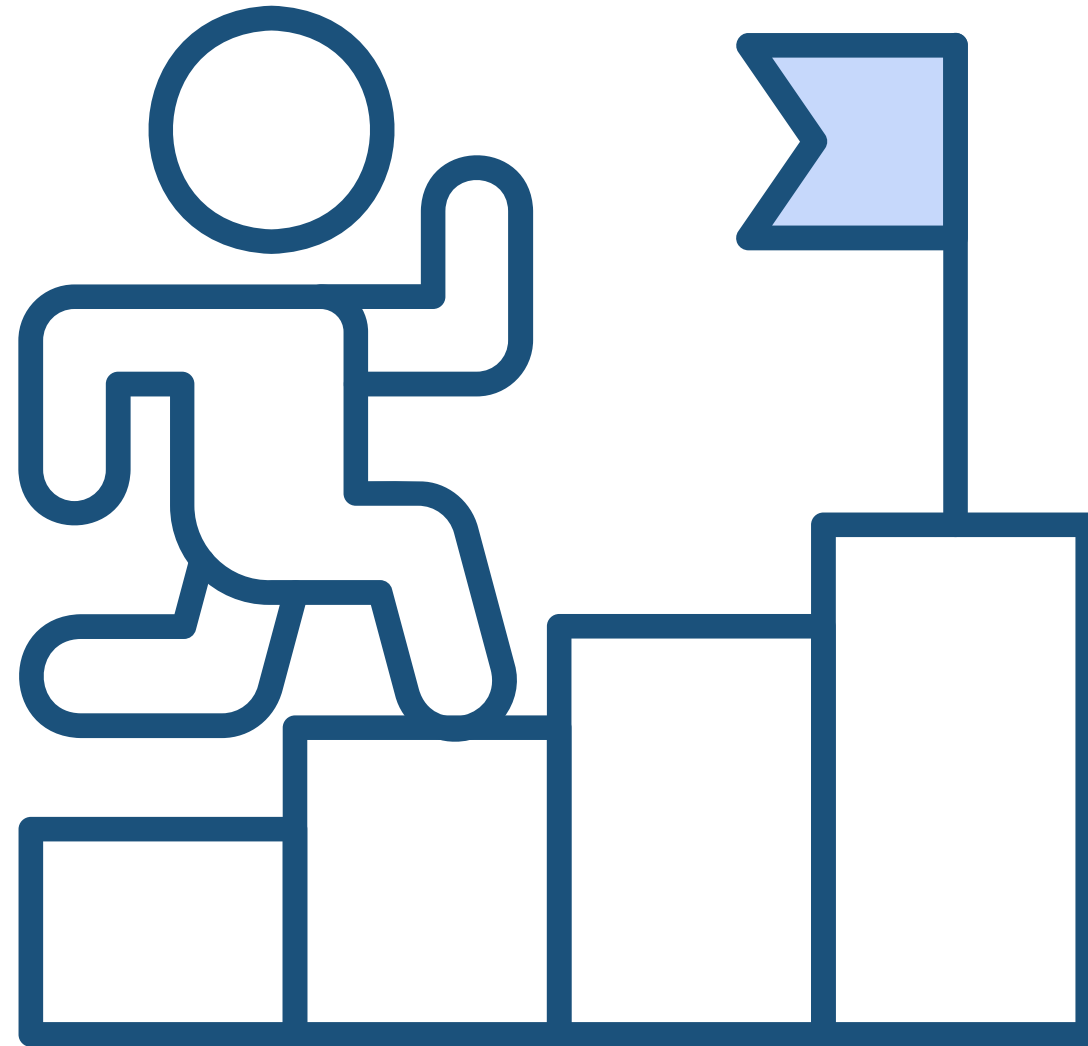


Triple burden of disease



Stepping up to this challenge

- Shifting paradigms from disease-based to person-centered care, moving to precision-focused models



- Developing meaningful partner with individuals, communities, funders and policy makers

- Nurse-led interventions can optimize outcomes of care and reduce the healthcare expenses



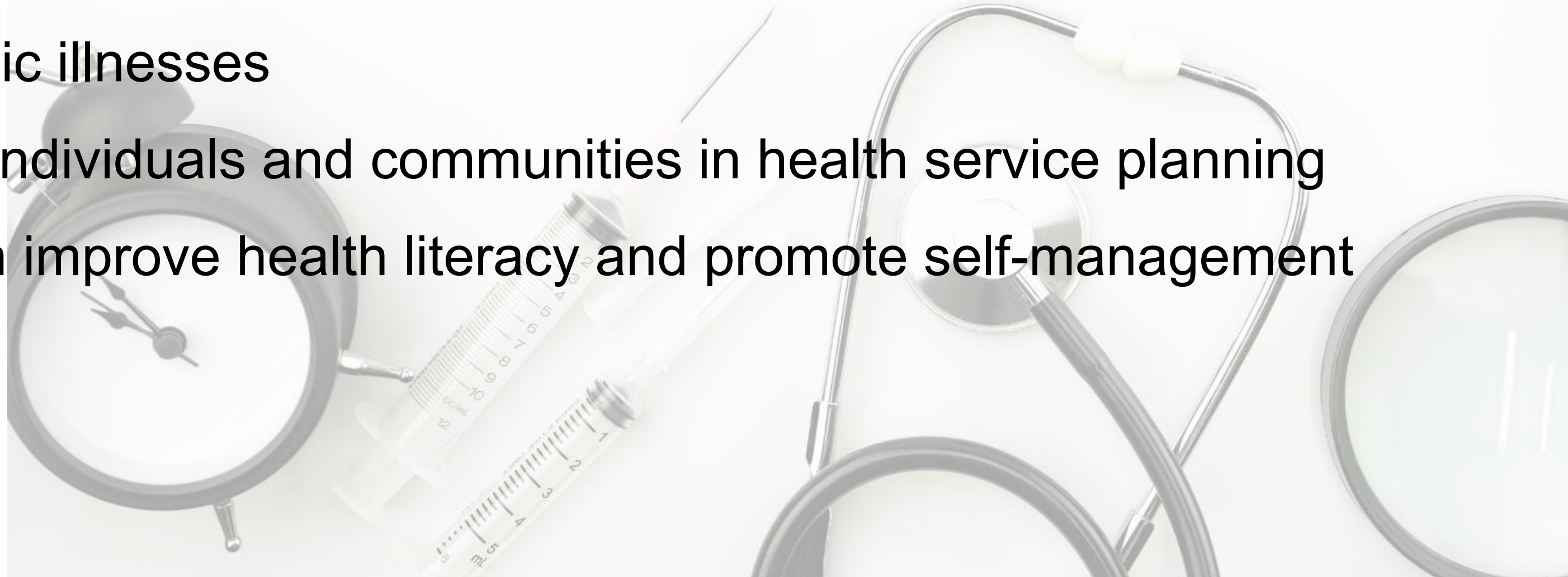
Promoting a healthy planet

- Developing strategies to mitigate the impact of climate change on health can help prevent the emergence of new diseases and reduce vulnerabilities in populations already facing multiple health challenges



Strengthening health and social care systems

- Investment in primary care and systems of universal health coverage is critical in the prevention, early diagnosis, and management of both acute and chronic illnesses
- Engaging with individuals and communities in health service planning and delivery can improve health literacy and promote self-management



Applying technology ethically and appropriately

- Applying telehealth approaches, mobile health applications, and electronic health records can enhance access to care, particularly in remote areas
 - >facilitate better communication between patients and providers
 - >improve access to service delivery
 - >enable monitoring of outcomes on an individual, organizational and systems level



A workforce for the future

- The ability of nurses and midwives to respond to current and future healthcare challenges
- Workforce maldistribution
- Poor working conditions
- Lack of a focus on education and leadership





Question And Answer Session

Thank You For Your Attention

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